



Additional Problems for Self Practice (APSP)

➤ Marked questions are recommended for Revision.

This Section is not meant for classroom discussion. It is being given to promote self-study and self testing amongst the Resonance students.

PART - I : PRACTICE TEST-1 (IIT-JEE (MAIN Pattern))

Max. Time : 1 Hr.

Max. Marks : 120

Important Instructions

- The test is of **1 hour** duration.
- The Test Booklet consists of **30** questions. The maximum marks are **120**.
- Each question is allotted **4 (four)** marks for correct response.
- Candidates will be awarded marks as stated above in Instructions No. 3 for correct response of each question. $\frac{3}{4}$ (**one fourth**) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 4 above.

- Which of the following has maximum ionisation energy ?
 (1) $\text{Ba} \rightarrow \text{Ba}^+ + e^-$ (2) $\text{Be} \rightarrow \text{Be}^+ + e^-$ (3) $\text{Ca} \rightarrow \text{Ca}^{2+} + 2e^-$ (4) $\text{Mg} \rightarrow \text{Mg}^{2+} + 2e^-$
- Which of the following increases in magnitude as the atomic number of alkali metals increases?
 (1) Electronegativity (2) First ionisation potential
 (3) Ionic radius (4) Melting point
- Alkali metals give colour in Bunsen flame due to :
 (1) Low electronegativity (2) One electron in outer most orbit
 (3) Smaller atomic radii (4) Low ionisation energy
- Which of the following s about solution of alkali metals in liquid ammonia is correct ?
 (1) The solutions have strong oxidizing properties.
 (2) Both the dilute solution as well as concentrated solution are paramagnetic in nature.
 (3) Colour of the solution is attributed to charge transfer spectrum.
 (4) None of these.
- Select the incorrect statement :
 (1) Solutions of alkali metals in liquid ammonia are a good reducing agents because they contain free or solvated electrons.
 (2) The crystalline salts of alkaline earth metals contain more water of crystallisation than the corresponding alkali metal salts.
 (3) Atoms of alkaline earth metals have smaller size and more nuclear charge than alkali metal atoms of same period.
 (4) All alkali metal halides form hydrates.
- Consider the following statements ;
S₁ : Among alkali metal halides lithium iodide is the most covalent in nature.
S₂ : Potassium has greater photoelectric work function than sodium.
S₃ : The blue solution of alkali metals in liquid ammonia is stable at room temperature, where ammonia is still a liquid, in the presence of Fe.
S₄ : The melting and boiling points of alkali metal halides always follow the trend : chloride > fluoride > bromide > iodide.
 and arrange in the order of true/false.
 (1) T F F F (2) T T F F (3) T F T T (4) T T T F
- Which of the following has the highest reactivity towards water ?
 (1) Na (2) Rb (3) Li (4) K
- Sodium burns in dry air to largely give :
 (1) Na_2O (2) Na_2O_2 (3) NaO_2 (4) Na_3N





9. Alkali metals are not characterised by :
 (1) good conductor of heat and electricity (2) high oxidation potentials
 (3) high melting points (4) solubility in liquid ammonia
10. In view of their ionisation energies, the alkali metals are :
 (1) weak oxidising agents (2) strong reducing agents
 (3) strong oxidising agents (4) weak reducing agents
11. Which of the following is incorrect statement :
 (1) Solubilities of alkaline earth metal fluorides and hydroxides generally increase down the group.
 (2) Hydration energies of alkali metal halides decrease down the group with increase in size of cations.
 (3) Mg^{2+} ion is bigger than Li^+ .
 (4) $BeCl_2$ is easily hydrolysed in water.
12. A metal (M) burns with dazzling brilliance in air above $1000^\circ C$ to give a white powder. The white powder reacts with water to form a white precipitate (P) and a colourless gas (G) with a characteristic smell. The metal (M) dissolves in conc. NaOH to liberate another gas(H). (H) may also be obtained on heating (G). Then :
 (1) $M = Mg$ (2) $M = Be$ (3) $P = Ca(OH)_2$ (4) $G = O_3$
13. Carbon can be oxidised to CO_2 while heating its powder with :
 (1) SO_2 (2) KNO_3 (3) K_2CO_3 (4) FeS_2
14. What happens when sodium metal is heated in excess of dry air containing carbon dioxide gas ?
 (1) Na_2O_2 is formed. (2) Na_2O is formed. (3) Na_2CO_3 is formed. (4) Na_3N is formed.
15. Solution of K_2O in water is basic, because it contains a significant concentration of :
 (1) O_2^{2-} (2) O_2^- (3) OH^- (4) K^+
16. Which of the following oxides is formed when potassium metal is burnt in excess air ?
 (1) KO_3 (2) K_2O (3) K_2O_2 (4) KO_2
17. On commercial scale, sodium hydroxide is prepared by :
 (1) Dow's process (2) Solvay process
 (3) Castner-Kellner cell (4) Hall-Heroult process
18. Which of the following gives sodium hydroxide along with hydrogen gas on reaction with water ?
 (1) Sodium oxide (2) Sodium amalgam (3) Sodium peroxide (4) Sodium carbonate.
19. Which of the following can exist in aqueous solution?
 (1) Na_2O (2) Na_2O_2 (3) KO_2 (4) K_2CO_3
20. Which of the following salts are composed of isoelectronic cation and anion
 (I) NaCl (II) $BaCl_2$ (III) MgF_2 (IV) CaS
 (1) I and II (2) II and III (3) III & IV (4) None of these
21. Which of the following liberates H_2 with cold water ?
 (1) H_2O_2 (2) NaH (3) NaOH (4) Mg
22. When ionic nitrides react with water, the products are :
 (1) acidic solution and hydrogen gas. (2) acidic solution and ammonia gas.
 (3) basic solution and ammonia gas. (4) basic solution and hydrogen gas.
23. Low solubility of CsI in water is due to :
 (1) smaller hydration enthalpy of Cs^+ . (2) smaller hydration enthalpy of I^- .
 (3) lower lattice enthalpy of its two ions. (4) (1) and (2) both.
24. Select the correct statement.
 (1) Among the alkali metals, only lithium reacts with nitrogen directly at room temperature to form nitride.
 (2) Among the alkali metal carbonates, Li_2CO_3 has the lowest thermal stability.
 (3) Among the alkali metal hydroxide, CsOH has the highest solubility in water.
 (4) All of these.
25. $NaNO_3$ is not used as gun powder because it is :
 (1) hygroscopic (2) very costly (3) amorphous (4) soluble in water



26. A doctor by mistake administers a dilute $\text{Ba}(\text{NO}_3)_2$ solution to a patient for radiographic investigations. Which of the following should be the best to prevent the absorption of soluble Barium and subsequent Barium poisoning.
 (1) NaCl (2) Na_2SO_4 (3) Na_2CO_3 (4) NH_4Cl
27. Baking powder used to make cake is a mixture of starch, NaHCO_3 and $\text{Ca}(\text{H}_2\text{PO}_4)_2$. The function of $\text{Ca}(\text{H}_2\text{PO}_4)_2$ is :
 (1) to slow down the release of CO_2 gas
 (2) it has acidic hydrogen and gives CO_2 when moistened with NaHCO_3
 (3) to act as a filler
 (4) None of these
28. Which salt hydrolyses to a minimum extent ?
 (1) $\text{Mg}(\text{NO}_3)_2$ (2) $\text{Be}(\text{NO}_3)_2$ (3) $\text{Ca}(\text{NO}_3)_2$ (4) $\text{Ba}(\text{NO}_3)_2$
29. Methanides are :
 (1) Mg_2C_3 , Be_2C , Al_4C_3 and CaC_2 (2) Mg_2C_3 , Be_2C and Al_4C_3
 (3) Be_2C , Al_4C_3 and CaC_2 (4) Be_2C and Al_4C_3
30. Select correct statement :
 (1) Interstitial carbides are formed by metalloids like Si and B.
 (2) SiC and B_4C are covalent carbides.
 (3) B_4C on hydrolysis gives methane.
 (4) VC, WC are ionic carbides.

Practice Test-1 (IIT-JEE (Main Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23	24	25	26	27	28	29	30
Ans.										

PART-II: NATIONAL STANDARD EXAMINATION IN CHEMISTRY (NSEC) STAGE-I

1. Which of the following is used in the photoelectric cells ? [NSEC-2000]
 (A) cesium (B) sodium (C) lithium (D) francium
2. Bleaching powder is : [NSEC-2000]
 (A) CaOCl_2 (B) CaClO (C) CaClO_3 (D) $\text{Ca}(\text{OCl})_2$
3. The chemistry of Li is very similar to that of Mg even though they belong to different groups. This is due to the fact that [NSEC-2000]
 (A) both occur in nature as compounds. (B) both have same electronic configuration
 (C) both have nearly the same size (D) both have charge to size ratio nearly the same.
4. Fire extinguisher contains H_2SO_4 and [NSEC-2000]
 (A) Na_2CO_3 (B) NaHCO_3 solution
 (C) NaHCO_3 & Na_2CO_3 (D) CaCO_3
5. Washing soda is manufactured by : [NSEC-2001]
 (A) Denny's process (B) Hall's process (C) Castner's process (D) Solvay's process
6. For which element crimson colour is obtained in flame test ? [NSEC-2002]
 (A) sodium (B) barium (C) strontium (D) calcium
7. Alkali metal dissolves in liquid ammonia at -33°C to produce [NSEC-2002]
 (A) violet colour (B) blue colour
 (C) reddish-violet colour (D) green colour.



8. Portland cement is [NSEC-2005]
 (A) calcium aluminium silicate (B) alumina
 (C) gypsum (D) calcium carbonate.
9. Crystals of washing soda lose nine molecules of water when exposed to dry air. This phenomenon is known as [NSEC-2005]
 (A) dehydration (B) efflorescence (C) deliquescence (D) evaporation.
10. Bleaching powder is made by passing Cl_2 into slaked lime. Its formula is [NSEC-2005]
 (A) $\text{Ca}(\text{OCl})_2$ (B) $\text{CaO}(\text{OCl})$ (C) CaOCl_2 (D) $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$.
11. The metal that shows photoelectric emission at lowest frequency radiation is [NSEC-2005]
 (A) beryllium (B) lithium (C) sodium (D) magnesium.
12. The lithium ion (Li^+) and hydride ion (H^-) are isoelectronic ions. Which statement about these systems is true ? [NSEC-2006]
 (A) Chemical properties of these ions are identical since they are isoelectronic.
 (B) Li^+ is a stronger reducing agent than H^-
 (C) More energy is needed to ionize H^- than Li^+
 (D) Radius of H^- is larger than that of Li^+ .
13. Which compound has largest lattice energy ? [NSEC-2006]
 (A) LiBr (B) LiCl (C) LiI (D) LiF .
14. Element having (4, 0, 0, $+1/2$) as a set of four quantum numbers for its valence electron is- [NSEC-2007]
 (A) Na (B) Ca (C) K (D) Br
15. The commercial name of calcium hydride is [NSEC-2012]
 (A) Lime (B) Hydrolyth (C) Slaked lime (D) Calgon
16. The crimson colour imparted to flame is due to a salt of : [NSEC-2012]
 (A) Barium (B) Copper (C) Calcium (D) Strontium
17. The chemical formula of Plaster of Paris is [NSEC-2013]
 (A) $2\text{CaSO}_4 \cdot \text{H}_2\text{O}$ (B) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (C) $3\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (D) $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
18. The correct statement for crystalline CsI_3 is [NSEC-2014]
 (A) it contains Cs^+ , I^- and molecular I_2 (B) it is a covalent compound
 (C) it contains Cs^+ and I_3^- (D) it contains Cs^{3+} and I^-
19. Sodium metal dissolves in liquid ammonia and forms a deep blue solution. The color is due to absorption of light by [NSEC-2015]
 (A) sodium ions (B) ammoniated electrons
 (C) free electrons (D) ammoniated sodium ions
- 20.* The reaction that is least feasible is [NSEC-2015]
 (A) $\text{Li}_2\text{CO}_3 \rightarrow \text{Li}_2\text{O} + \text{CO}_2$ (B) $4\text{Li} + \text{O}_2 \rightarrow 2\text{Li}_2\text{O}$
 (C) $6\text{Li} + \text{N}_2 \rightarrow 2\text{Li}_3\text{N}$ (D) $2\text{C}_6\text{H}_5\text{C}\equiv\text{CH} + 2\text{Li} \rightarrow 2\text{C}_6\text{H}_5\text{C}\equiv\text{CLi} + \text{H}_2$
21. A dilute solution of an alkali metal in liquid ammonia is [NSEC-2018]
 I. blue in colour II. conducts electricity III. paramagnetic IV. an oxidizing agent
 (A) I and III (B) II and IV (C) I, II and III (D) I and III

PART - III : PRACTICE TEST-2 (IIT-JEE (ADVANCED Pattern))

Max. Time : 1 Hr.

Max. Marks : 69

Important Instructions

A. General :

- The test is of 1 hour duration.
- The Test Booklet consists of 23 questions. The maximum marks are 69.

B. Question Paper Format

- Each part consists of five sections.
- Section-1 contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE is correct.



5. Section-2 contains 6 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE OR MORE THAN ONE are correct.
6. Section-3 contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9 (both inclusive).
7. Section-4 contains 1 paragraphs each describing theory, experiment and data etc. 3 questions relate to paragraph. Each question pertaining to a particular passage should have only one correct answer among the four given choices (A), (B), (C) and (D).
8. Section-5 contains 1 multiple choice questions. Question has two lists (list-1 : P, Q, R and S; List-2 : 1, 2, 3 and 4). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

C. Marking Scheme

9. For each question in Section 1, 4 and 5 you will be awarded 3 marks if you darken the bubble corresponding to the correct answer and zero mark if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.
10. For each question in Section 2, you will be awarded 3 marks. If you darken all the bubble(s) corresponding to the correct answer(s) and zero mark. If no bubbles are darkened. No negative marks will be answered for incorrect answer in this section.
11. For each question in Section 3, you will be awarded 3 marks if you darken only the bubble corresponding to the correct answer and zero mark if no bubble is darkened. No negative marks will be awarded for incorrect answer in this section.

SECTION-1 : (Only One option correct Type)

This section contains 7 Single correct questions. Each questions has four choices (A), (B), (C) and (D) out of which Only ONE option is correct.

1. Consider the following statements :
 S_1 : Beryllium and Magnesium are inert to oxygen and water.
 S_2 : Concentrated solutions of alkaline earth metals in ammonia are bronze coloured.
 S_3 : Calcium, strontium and barium reacts with cold water forming hydroxides and liberating hydrogen gas.
 S_4 : Oxides and hydroxides of alkaline earth metals are more ionic and more basic than that of the alkali metals.
 and arrange in the order of true/false.
 (A) T T T T (B) T T T F (C) F T T F (D) F T F F
2. Which of the following statement is incorrect ?
 (A) The superoxide ion (i.e., O_2^-) is stable only in presence of larger cations such as K, Rb, Cs.
 (B) Alkali metals are normally kept in kerosene oil.
 (C) All the alkali metal hydrides are ionic solids with high melting points.
 (D) The concentrated solution of alkali metals in liquid ammonia are strong paramagnetic in nature.
3. Sodium is heated in excess of air, free from CO_2 at $350^\circ C$ to form X. X absorbs CO_2 and form Na_2CO_3 and Y. 'X' and 'Y' are respectively :
 (A) Na_2O and O_2 (B) Na_2O_2 and O_2 (C) NaO_2 and O_2 (D) Na_2O_2 and O_3
4. What products are formed during the electrolysis of a concentrated aqueous solution of sodium chloride?
 I. $Cl_2(g)$, II. $NaOH(aq)$, III. $H_2(g)$.
 (A) I only (B) I and II only (C) I and III only (D) All of these
5. The following flow diagram represents the manufacturing of sodium carbonate ?
 $2 NH_3 + H_2O + CO_2 \longrightarrow (NH_4)_2CO_3 \xrightarrow{(a)} NH_4HCO_3 \xrightarrow{(b)} NaHCO_3 + (c)$
 $\downarrow (d)$
 $Na_2CO_3 + CO_2 + H_2O$

Which of the following option describes the underlined reagents, products and reaction conditions ?

Option	(a)	(b)	(c)	(d)
(A)	Carbon dioxide	NaCl	NH_4Cl	Heat
(B)	Carbon dioxide	NaCl	NH_4Cl	catalyst
(C)	Higher tempt.	NaCl	NH_4Cl	Heat
(D)	Higher pressure	NaCl	NH_4Cl	Catalyst



6. Which of the following statement is incorrect ?
 (A) The effective component of bleaching powder is OCl^- .
 (B) CaCO_3 is obtained when quick lime is heated with coke in an electric furnace.
 (C) Anhydrous CaSO_4 is dead burnt plaster.
 (D) BaCO_3 is obtained on fusion of BaSO_4 and Na_2CO_3 .
7. Consider the following statements,
 S_1 : Gypsum contains a lower percentage of calcium than plaster of paris.
 S_2 : Plaster of paris can be re-obtained by hydration of 'dead plaster'.
 S_3 : Gypsum loses $3/2$ of its water of crystallisation forming plaster of paris at 120°C .
 S_4 : Plaster of paris can be obtained by partial oxidation of gypsum.
 and arrange in the order of true/false.
 (A) T F T F (B) F F T F (C) T T F F (D) T T T T

Section-2 : (One or More than one options correct Type)

This section contains 6 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE THAN ONE are correct.

8. Select the correct statement with respect to the deep blue solution of an alkali metal in liquid ammonia.
 (A) Its paramagnetism decreases with increasing concentration.
 (B) It has lower density than pure solvent (i.e. liquid ammonia).
 (C) Its conductivity decreases with increasing concentration to minimum at about 0.05 molar; thereafter it again increases.
 (D) Evaporation of ammonia from the solution yields alkali metal.
9. Freshly prepared pure dilute solution of sodium in liquid ammonia :
 (A) shows copper - bronze colour.
 (B) occupy larger volume than that from the sum of the volumes of Na and $\text{NH}_3(l)$.
 (C) reduces the GeH_4 to GeH_3^- .
 (D) produces sodium amide and hydrogen gas with rusty iron wire.
10. Which of the following disproportionate(s) on heating with sodium hydroxide ?
 (A) P_4 (white) (B) S_8 (C) Cl_2 (D) B
11. Which of the following statement(s) is/are correct ?
 (A) Pure sodium oxide is obtained by heating the mixture of sodium azide and sodium nitrite.
 (B) Glauber's salt effloresces in moist air.
 (C) Potassium superoxide on heating with sulphur in an evacuated and sealed tube yields potassium thiosulphate.
 (D) Gypsum dissolve in ammonium sulphate solution.
12. Select correct statement(s) :
 (A) CaCO_3 is more soluble in a solution of CO_2 than in H_2O .
 (B) Na_2CO_3 is converted to Na_2O and CO_2 on heating.
 (C) Li_2CO_3 is thermally unstable.
 (D) Presence of CaCl_2 or CaSO_4 in water causes temporary hardness.
13. The pair (s) of compounds which can exist together in aqueous solutions is/are ?
 (A) NaH_2PO_4 and Na_2HPO_4 (B) NaHCO_3 and NaOH .
 (C) Na_2HPO_3 and NaOH (D) NaHSO_4 and NaOH .

Section-3 : (One Integer Value Correct Type.)

This section contains 6 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive)

14. $\text{M} + (\text{X} + \text{Y})\text{NH}_3 \longrightarrow [\text{M}(\text{NH}_3)_x]^{2+} + 2[\text{e}(\text{NH}_3)_4]^-$; where M = alkaline earth metal.
 Value of x is....
15. How many of the following are correctly matched :
 Hydration energy : $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$
 Mobility of ions (aq) : $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$
 Density : $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$
 Reaction with N_2 : $\text{Li} < \text{Na} < \text{Rb} < \text{Cs} < \text{K}$
 Reducing nature of gas phase : $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$



Reducing nature in aq. phase : $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$

16. How many of the following statements are correct ?
 (a) BeO is amphoteric in nature.
 (b) LiHCO_3 is not found in solid state.
 (c) K_2O_2 is diamagnetic but KO_2 is paramagnetic.
 (d) White phosphorous react with caustic soda and gives phosphine gas.
 (e) AlCl_3 is soluble in excess of NaOH and form sodium meta aluminate.
 (f) Anhydrous potassium nitrate on heating with potassium metal gives potassium oxide and nitrogen gas.
 (g) Lithium chloride is highly soluble in water.
 (h) Hydrated magnesium chloride on heating in dry air gives anhydrous MgCl_2 .
17. How many of the following orders are correct :
 (A) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Ba}(\text{OH})_2$ Basic character
 (B) $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$ Decomposition temperature
 (C) $\text{Na}^+ > \text{Mg}^{2+} > \text{Li}^+ > \text{Be}^{2+}$ Size
 (D) $\text{Li}_2\text{CO}_3 > \text{Na}_2\text{CO}_3 > \text{K}_2\text{CO}_3 > \text{Rb}_2\text{CO}_3 > \text{Cs}_2\text{CO}_3$ Water solubility
 (E) $\text{Na}_2\text{O}_2 < \text{K}_2\text{O}_2 < \text{Rb}_2\text{O}_2 < \text{Cs}_2\text{O}_2$ Stability
 (F) $\text{LiHCO}_3 < \text{NaHCO}_3 < \text{KHCO}_3 < \text{RbHCO}_3 < \text{CsHCO}_3$ Stability
 (G) $\text{NaF} < \text{NaCl} < \text{NaBr} < \text{NaI}$ Melting point
 (H) $\text{Na}_2\text{O}_2 < \text{KO}_2 < \text{O}_2[\text{AsF}_4]$ O—O bond length
18. How many of the following bicarbonates are solid in nature?
 LiHCO_3 , NaHCO_3 , KHCO_3 , RbHCO_3 , CsHCO_3 , $\text{Be}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$, $\text{Ca}(\text{HCO}_3)_2$, $\text{Sr}(\text{HCO}_3)_2$, $\text{Ba}(\text{HCO}_3)_2$
19. How many types of products are formed when LiNO_3 and NaNO_3 are heated at 500°C

SECTION-4 : Comprehension Type (Only One options correct)

This section contains 1 paragraph, describing theory, experiments, data etc. 3 questions relate to the paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 20 to 22

Alkali metals oxide are obtained by combustion of the metals. Although Na normally gives Na_2O , it will take up further oxygen at elevated pressure and temperatures to form NaO_2 . The per and superoxides of the heavier alkalies can also be prepared by passing stoichiometric amounts of oxygen into their solution in liquid ammonia.

The different alkali metal oxides can be distinguished by reaction with water. The superoxides reacts with CO_2 and give oxygen gas. The stability of per and superoxides is based upon that larger cation can stabilise larger anion, due to larger lattice energy.

Alkali metals dissolve in liquid ammonia. Dilute solutions are dark blue in colour but as the concentration increases above 3M, the colour changes to copper bronze and the solution acquires the metallic lustre due to the formation of metal ions clusters. The solution of alkali metals in liquid ammonia are good conductors of electricity due to the presence of ammoniated cations and ammoniated electrons. However, the conductivity decreases as the concentrations increases, since ammoniated electrons and ammoniated cation associate.

20. Solution of sodium metals in liquid ammonia is strongly reducing due to the presence of :
 (A) Sodium hydride (B) Sodium atoms (C) Sodium amide (D) Solvated electrons.
21. KO_2 is used in oxygen cylinders in space and submarines because it.
 (A) Eliminates moisture (B) Absorbs CO_2 only
 (C) Absorbs CO_2 and increases O_2 contents (D) Produces ozone.
22. Select the correct choice for alkali metal oxides.
 (A) Metal oxides reacts with water forming only metal hydroxides
 (B) Metal peroxides reacts with warm water forming metal hydroxides and oxygen gas
 (C) Metal superoxides reacts with water forming metal hydroxide, Hydrogen peroxide and O_2 gas
 (D) All of these



SECTION-5 : Matching List Type (Only One options correct)

This section contains 1 questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D) out of which one is correct

23. Match the reactions given in **List I** with the main products obtained and given in **List II** and select the correct answer using the code given below the lists.

	List - I (Reaction at given temperature)		List - II (Reaction involve)
(P)	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{200^\circ\text{C}, \Delta}$	(1)	Formation of lime
(Q)	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{1100^\circ\text{C}, \Delta}$	(2)	Setting of plaster of paris
(R)	$2\text{CaSO}_4 \cdot \text{H}_2\text{O} + \text{H}_2\text{O} \xrightarrow{\text{room temp.}}$	(3)	Formation of burnt plaster
(S)	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{120^\circ\text{C}, \Delta}$	(4)	Formation of plaster of paris

Code :

	P	Q	R	S		P	Q	R	S
(A)	1	3	4	2	(B)	4	1	2	3
(C)	3	1	2	4	(D)	4	2	1	3

Practice Test-2 (IIT-JEE (ADVANCED Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23							
Ans.										

APSP Answers

PART - I

1.	(4)	2.	(3)	3.	(4)	4.	(4)	5.	(4)
6.	(1)	7.	(2)	8.	(2)	9.	(3)	10.	(2)
11.	(3)	12.	(2)	13.	(2)	14.	(3)	15.	(3)
16.	(4)	17.	(3)	18.	(2)	19.	(4)	20.	(3)
21.	(2)	22.	(3)	23.	(4)	24.	(4)	25.	(1)
26.	(2)	27.	(2)	28.	(4)	29.	(4)	30.	(2)

PART-II

1.	(A)	2.	(A)	3.	(D)	4.	(C)	5.	(D)
6.	(C)	7.	(B)	8.	(A)	9.	(B)	10.	(C)
11.	(C)	12.	(D)	13.	(D)	14.	(C)	15.	(B)
16.	(D)	17.	(A)	18.	(C)	19.	(B)	20.	(AD)
21.	(C)								



PART - III

1. (B)	2. (D)	3. (B)	4. (D)	5. (A)
6. (B)	7. (A)	8. (ABCD)	9. (BCD)	10. (ABC)
11. (AD)	12. (AC)	13. (AC)	14. 6	15. 2
16. 6.	17. 5 (Except D,G,H)		18. 4	19. 4
20. (D)	21. (C)	22. (D)	23. (C)	

APSP Solutions

PART - I

- Down the group size increases and therefore, attraction between valence shell electron and nucleus decreases and thus ionisation energy decreases.
- Down the group, the atomic size increases with increasing atomic number and so attraction for shared pair of electrons decreases. Hence electronegativity decreases.
 - Down the group, the atomic size increases with increasing atomic number and so attraction between valence electron and nucleus decreases. Hence ionization energy decreases.
 - Down the group atomic size increases with increase in number of atomic shells while effective nuclear charge remains constant.
 - Down the group atomic size increases with increase in number of atomic shells and therefore, the strength of metallic bond decreases. So melting point decreases.
- Alkali metals have low ionisation energy.
- All are wrong. The solution has strong reducing nature and coloured due to ammoniated electron. Dilute solution is paramagnetic whereas concentrated solution is diamagnetic.
- Smaller cation and higher charge attracts more numbers of water molecules.
 - Periodic property
 - Except Li^+ , due to bigger size of ions they have low hydration enthalpies. Hence except lithium, all alkali metal halides do not form hydrates.
- S₁** : Li^+ being smaller have high polarising power and I^- being larger have high polarisability. So it is most covalent among alkali metal halides according to Fajan's rule.

S₂ : The IE₁ of potassium atom is less than sodium atom.

S₃ : The presence of transition metals like iron and other impurities catalyses the decomposition of deep blue solution forming amide and liberating H_2 .

S₄ : Two opposing tendencies exist. With greater charge and smaller size of cation, lattice energy increases which tends to increase the melting point; while increase in covalent character causes a decrease in melting point. Hence, no unique generalised trend may be stated for melting points. (Students need not worry about or memorise such experimental data).
- The reaction of alkali metals with water becomes increasingly violent on descending the group on account of their decreasing ionisation energies with increasing atomic size. So, the order of reactivity is: $\text{Li} < \text{Na} < \text{K} < \text{Rb}$.
- $2\text{Na} + \text{O}_2 \longrightarrow \text{Na}_2\text{O}_2$
- They have weak metallic bond because of one valence electron per atom. So they have low melting points.
- They easily lose valence shell electron because of their low ionisation energies, on account of their bigger atomic sizes. So they behave as strong reducing agents.
- Factual
 - Hydration energy $\propto \frac{1}{\text{size of cation}}$.
 - Both are diagonally related; because of more positive charge on Mg, Mg^{2+} is smaller than Li^+ .
 $\text{Li}^+ = 76 \text{ pm}$, $\text{Mg}^{2+} = 72 \text{ pm}$.
 - Salt of weak base and strong acid, thus easily hydrolysed in water giving acidic solution.



12. $\text{Be} + \text{Air} \xrightarrow{T > 1000^\circ\text{C}} \text{BeO} + \text{Be}_3\text{N}_2$ (white powder)
 $\text{Be}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Be}(\text{OH})_2$ (white precipitate) + 2NH_3 (Colourless gas)
13. $\text{KNO}_3 \xrightarrow{\text{Heat}} \text{KNO}_2 + \frac{1}{2} \text{O}_2$
 $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
14. $2\text{Na} + \text{O}_2 \xrightarrow{\text{Heat}} \text{Na}_2\text{O}_2$; $2\text{Na}_2\text{O}_2 + 2\text{CO}_2 \rightarrow 2\text{Na}_2\text{CO}_3 + \text{O}_2$
15. $\text{K}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{KOH} \rightarrow \text{K}^+ + \text{OH}^-$.
 The resulting solution is basic due to the presence of OH^- .
16. $\text{K} + \text{O}_2 \xrightarrow{\text{burning}} \text{KO}_2$.
17. Sodium hydroxide is manufactured by the electrolysis of brine using Castner-Kellner cell.
18. (1) $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$
 (2) $2\text{Na/Hg} + 2\text{H}_2\text{O}$ (Castner-Kellner cell) $\rightarrow 2\text{NaOH} + 2\text{Hg} + \text{H}_2$.
 (3) $\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$
 (4) $\text{Na}_2\text{CO}_3 + 2\text{H}_2\text{O} \rightleftharpoons 2\text{NaOH} + \text{H}_2\text{CO}_3$
19. (1), (2) & (3) reacts with water being more basic than water.
20. Isoelectronic species have same number of electrons.
21. All alkali metal hydrides are ionic in nature and react with water according to the reaction ;
 $\text{NaH} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2$.
22. $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$
23. True statement. The CsI, because of bigger cation (Cs^+) and bigger anion (I^-), has smaller hydration enthalpy. As a result, it does not exceed its lattice energy; so CsI is insoluble in water.
24. (1) Lithium show exceptional behavior in reaction directly with nitrogen of air to form the nitride, Li_3N .
 (2) Smaller cation (Li^+) polarises bigger anion (Cl_3^{2-}) liberating CO_2 gas. So it has the lowest thermal stability.
 (3) The solubility of the alkali metal hydroxides increases down the group from Li to Cs. This is because of the fact that down the group with increasing size of cation, the lattice energy as well as hydration energy also decreases but the change in lattice energy is more as compare to that of hydration energy.
25. NaNO_3 is not used as gun powder because it is hygroscopic in nature and becomes wet by absorbing water molecules from the atmosphere. Therefore, (1) option is correct.
26. Na_2SO_4 .
27. Baking powder used to make cake is a mixture of starch, NaHCO_3 and $\text{Ca}(\text{H}_2\text{PO}_4)_2$. The function of $\text{Ca}(\text{H}_2\text{PO}_4)_2$ is being acidic in nature and gives CO_2 when moistened with NaHCO_3 .
28. $\text{Ba}(\text{NO}_3)_2$ results a neutral solution as it is the salt of strong acid, HNO_3 and strong base, $\text{Ba}(\text{OH})_2$.
29. Methanides give CH_4 on reaction with H_2O .
 $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \rightarrow 4\text{Al}(\text{OH})_3 + 3\text{CH}_4$; $\text{Be}_2\text{C} + 4\text{H}_2\text{O} \rightarrow 2\text{Be}(\text{OH})_2 + \text{CH}_4$
30. Factual.

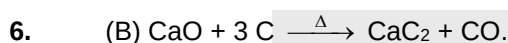
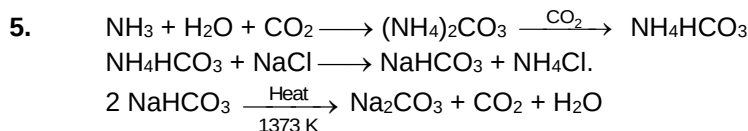
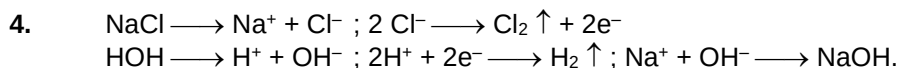
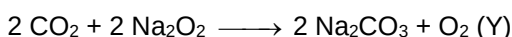
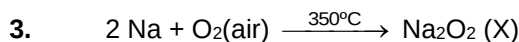
PART - III

1. **S₁** : Because of the formation of an oxide film on their surface.
S₂ : Due to the formation of metal ion clusters.
S₃ : As the basicity (i.e. electropositive character) of alkaline earth metals increases, their reactivity towards water increases.
S₄ : Oxides and hydroxides of alkaline earth metals are less ionic and basic. This is due to increased nuclear charge and smaller size.
2. (A) Bigger anion is stabilised by bigger cation through lattice energy effect.
 (B) Because of their high reactivity towards air and water on account of their higher electropositive character



(C) All alkali metals are highly basic in nature and, therefore, their hydrides are ionic solids with high melting points.

(D) In concentrated solution, unpaired electrons with opposite spins paired up-forming the solution diamagnetic.



8. (A) Very dilute solutions of the metals are paramagnetic, with approximately one unpaired electron per metal atom (corresponding to one solvated electron per metal atom); this paramagnetism decreases at higher concentration because of the association of unpaired electrons of opposite spins.

(B) According to the cavity concept, they occupy far greater volume than that expected from the sum of the volumes of metal and solvent and, therefore, these dilute solutions are of much lower density than pure solvent.

(C) Conduction is due mainly to the presence of solvated electrons. As the solutions are made more concentrated, the molar conductivity at first decreases, reaching a minimum at about 0.05 molar; thereafter, it increases again until in saturated solutions (it is comparable to that of the metal).

(D) Evaporation of the ammonia from solutions of alkali metals yields the metal, but with alkaline earth metals evaporation of ammonia gives hexamminates of the metals, $[\text{M}(\text{NH}_3)_6]^{2+}$.

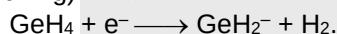
9. (A) The alkali metals dissolve in liquid ammonia giving blue solutions.



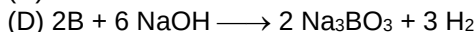
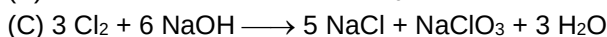
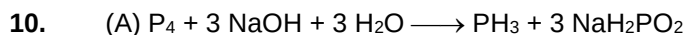
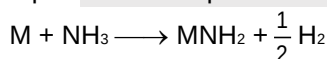
The blue colour, corresponding to a broad absorption band near 1500 nm that tails into the visible range, is attributed to the solvated electron.

(B) According to the cavity concept, they occupy far greater volume than that expected from the sum of the volumes of metal and solvent.

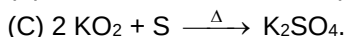
(C) The solution of metals in liquid ammonia act as powerful reducing agents (it even reduces the aromatic ring)



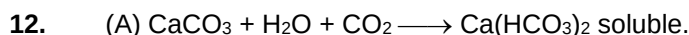
(D) In the presence of impurities or transition metals like Fe; MNH_2 and H_2 gas are formed.



(B) False $\rightarrow$ Glauber's salt ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$) effloresces in dry air.



(D) Form soluble double sulphate, $(\text{NH}_4)_2\text{SO}_4 \cdot \text{CaSO}_4 \cdot \text{H}_2\text{O}$ (soluble complex).



(B) Sodium is more basic (i.e. more ionic) in nature; so Na_2CO_3 is thermally stable towards heat. It does not decompose to give Na_2O and CO_2



(C) Li is least basic (i.e. more covalent) in nature; so Li_2CO_3 is thermally unstable.

(D) Presence of CaCl_2 or CaSO_4 in water causes permanent hardness. Temporary hardness of water is due to the presence of bicarbonates of Ca^{2+} and Mg^{2+} .

13. (A) Both are acid salts; so they can exist together in aqueous solution.

(B), (D) A base (NaOH) and an acid salt (NaHSO_4 and NaHCO_3) cannot exist together in solution.

Acid + base $\longrightarrow$ salt + water.

(C) Na_2HPO_3 is a neutral salt; so it does not further react with NaOH .

14. When alkaline earth metal except Be, Mg is dissolved in liquid NH_3 , we get hexaammoniated metal ion.

15. Hydration energy : $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$

Mobility of ions (aq) : $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$

Density : $\text{Li} < \text{K} < \text{Na} < \text{Rb} < \text{Cs}$

Reaction with N_2 : $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$

Reducing nature of gas phase : $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$

Reducing nature in aq. phase : $\text{Li} > \text{Cs} > \text{Rb} > \text{K} > \text{Na}$

16. (a), (b), (c), (d), (e) & (f) are correct.

(a) BeO is amphoteric in nature because it reacts with acid as well as base.

(b) $\text{LiHCO}_3 \xrightarrow{\Delta} \text{Li}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$

(d) $\text{H}_2\text{O} + \text{NaOH} + \text{P}_4 \longrightarrow \text{NaH}_2\text{PO}_2 + \text{PH}_3$

(e) $\text{AlCl}_3 + 4\text{NaOH (excess)} \longrightarrow \text{NaAlO}_2 + 2\text{H}_2\text{O} + 3\text{NaCl}$

(f) $\text{KNO}_3 + \text{K} \longrightarrow \text{K}_2\text{O} + \text{N}_2$

(h) $\text{MgCl}_2 \cdot 6\text{H}_2\text{O} \longrightarrow \text{MgO} + \text{HCl} + \text{H}_2\text{O}$.

17. 5 (Except D, G, H)

18. Only NaHCO_3 , KHCO_3 , RbHCO_3 , CsHCO_3 are present in solid form.

19. $\text{LiNO}_3 \xrightarrow{\Delta} \text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$

$\text{NaNO}_3 \xrightarrow{\Delta} \text{NaNO}_2 + \text{O}_2$

20. Here solvated electrons acts as a reducing agent.

21. $4\text{KO}_2 + 3\text{CO}_2 \longrightarrow 2\text{K}_2\text{CO}_3 + 3\text{O}_2$

22. (A) $\text{M}_2\text{O} + \text{H}_2\text{O} \longrightarrow 2\text{MOH}$

(B) $\text{M}_2\text{O}_2 + \text{H}_2\text{O} \longrightarrow 2\text{MOH} + \frac{1}{2}\text{O}_2$

(C) $2\text{MO}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{MOH} + \text{H}_2\text{O}_2 + \text{O}_2$

23. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{120^\circ\text{C}, \Delta} \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} \xrightarrow{200^\circ\text{C}, \Delta} \text{CaSO}_4 \xrightarrow{1100^\circ\text{C}, \Delta} \text{CaO} + \text{SO}_3$

(Gypsum)

(plaster of paris)

(burnt or dead plaster)

(lime)